SWARUP KUMAR SUBUDHI

Ph.D., Mechanical Engineering · University of Maryland, College Park
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RESEARCH INTERESTS

Printed and flexible electronics · Additive manufacturing for electronics and structural applications · Soft magnetic nanocomposites · Reliability engineering and physics of failure · Interfacial and magnetohydrodynamic fluid mechanics · Wearable and lab-on-chip sensing technologies · Advanced materials characterization · Process qualification for space and defense hardware

EDUCATION

Ph.D., Mechanical Engineering — 2026

University of Maryland, College Park, MD, USA · GPA: 3.7 / 4.0
Dissertation: Magnetic Films & Drops, and Their Interactions with Soft Materials
Advisor: Prof. Siddhartha Das, Department of Mechanical Engineering

Key Courses: Probabilistic Physics of Failure & Accelerated Testing, Failure Mechanisms of Electronic Packages, Composite Materials, Modelling Material Behavior, Mechanical Fundamentals of Electronic Components, Interfacial Fluid Mechanics

M.S., Mechanical Engineering — 2024

University of Maryland, College Park, MD, USA
Graduate Teaching Assistant: Fluid Mechanics (ENME331), Engineering Mechanics I (ENES102)

B.Tech., Mechanical Engineering — 2016

National Institute of Technology (NIT), Rourkela, Odisha, India
Thesis: Synthesis and Analysis of a Reconfigurable Manipulator Assembly Based on Five-Bar Linkage

Key Courses: Kinematics & Dynamics of Machinery, Machine Design I & II, Tool Design, Mechatronics, Digital Logic Design, Materials Technology, Mathematics (Minor)

RESEARCH EXPERIENCE

Graduate Research Assistant — Jan 2022 – Present

Department of Mechanical Engineering, University of Maryland, College Park, MD

Doctoral Dissertation — Magnetic Films & Drops, and Their Interactions with Soft Materials

PI & Sponsor: Dr Siddhartha Das, Jan 2022 – Dec 2026

- Experimentally demonstrated magnetohydrodynamic de-encapsulation of superparamagnetic nanoparticle (MNP) aggregates from the interior of a sessile droplet using external magnetic fields; characterized aggregate morphology, size distribution, and extraction kinetics
- Demonstrated selective cargo transfer of MNP aggregates and encapsulated payload between adjacent droplets without physical contact — a key enabler for open-microfluidic lab-on-chip drug delivery platforms
- Fabricated free-standing PDMS–Fe_3O_4 soft magnetic nanocomposite films: synthesized and characterised oleic-acid-coated Fe_3O_4 MNPs (TEM, zeta-potential, VSM), dispersed in PDMS matrix, and characterized magnetic properties (VSM, hysteresis), mechanical flexibility (DMA), and RF/EMF shielding performance
- Investigated wetting dynamics and contact-angle evolution of MNP-laden drops on hydrophobic/hydrophilic surfaces as a function of MNP concentration, magnetic Bond number, and external field orientation
- Developed Python (OpenCV/CV2) image processing algorithms for automated droplet tracking, aggregate volume estimation from high-frame-rate videos, and point-cloud reconstruction of MNP aggregate 3D morphology
- Studied drying dynamics of particle-laden drops: ring-stain formation, Marangoni flows, and 3D deposit morphology as functions of particle concentration and substrate wettability (in collaboration, published Langmuir 2023)

Project 1: Development of Additively Manufactured High-Energy X-Ray Photoelectric Polarimeter

Sponsor: NASA-Goddard Space Flight Center · Jan 2024 – Dec 2026
PI: Dr. Siddhartha Das (UMD), Dr. Margaret Samuels (GSFC) & Dr. Abhijit Dasgupta (UMD)

- Achieved ±10 μm print tolerances for high-density (50–100 μm pitch) silver-nanoflake interconnects on space-grade substrates by direct-write extrusion printing, enabling a compact X-ray polarimeter assembly qualified for launch environments
- Developed novel extrusion toolpaths for printing on irregular and curved substrate geometries; systematically tuned ink rheology (viscosity, surface tension, particle size distribution) to eliminate satellite drops and bridging defects
- Designed and executed vibration qualification test matrices (launch-load profiles per NASA GEVS) for printed specimens; established pass/fail criteria and authored qualification documentation for program reviews
- Conducted environmental reliability tests (thermal exposure, humidity) on printed interconnects; characterized resistance drift and adhesion degradation as functions of thermal history
- Performed dimensional metrology (optical microscopy, SEM, surface profilometry) to verify print fidelity and cross-section quality of multi-layer silver and dielectric stacks

Project 2: Thermal Cycling Reliability of Aerosol-Jet and Extrusion-Printed Electronics

Sponsor: Laboratory of Physical Sciences, June 2023 – Dec 2023

- Built parametric multi-layer finite element models (ANSYS Mechanical, Abaqus CAE) of printed dielectric (NEA121, BCB) and conductive (sintered silver) stacks under thermal cycling (−40 °C to +140 °C, JEDEC JESD22-A104)
- Identified failure initiation sites via inelastic strain accumulation (Anand viscoplasticity model) and cyclic creep in sintered silver; predicted crack propagation paths in agreement with physical cross-sections
- Validated FEA predictions against experimental failure analysis: SEM imaging, EDS elemental mapping, and optical cross-section microscopy of cycled specimens
- Characterized material parameters (CTE, elastic modulus, creep coefficients) using TMA, DMA, and DSC for accurate model inputs

Project 3: Reliability Validation of Direct-Write Printed RF Devices

Sponsor: NextFlex in collab with Binghampton University & Lockheed Martin, Jan 2022 – May 2023

- Fabricated Metal-Insulator-Semiconductor (MIS) capacitors and coplanar waveguides (CPW) by extrusion direct-write printing of dielectric (NEA121) and silver paste on flat and curved substrates
- Developed custom print-head trajectory algorithms to achieve conformal deposition on non-planar surfaces; demonstrated functional RF prints on cylindrical and doubly-curved geometries
- Characterized RF performance using vector network analyzer (VNA): S-parameters (S11, S21), insertion loss, reflection coefficients, and quality factor Q across 0.1–6 GHz
- Qualified specimens against program specs for RF performance retention after thermal and mechanical stress; documented failure modes and process improvement recommendations

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Graduate Teaching Assistant

Jan 2021 – Dec 2021

Department of Mechanical Engineering, University of Maryland, College Park, MD

- Instructed lab and studio sessions Fluid Mechanics (ENME331) and Engineering Mechanics I (ENES102)
- Held weekly office hours; graded assignments, quizzes, and exams for classes of 80+ students
- Developed supplementary worked-example problem sets for fluid statics, pipe flow, and rigid body dynamics

PROFESSIONAL EXPERIENCE

Senior Manager, Manufacturing — Transmissions
2019 – 2020

Tata Motors Ltd., CVBU, Jamshedpur, Jharkhand, India

Six Sigma: Warranty Cost Reduction — Gearbox Bearing Failures

- Led a cross-functional Six Sigma project (NPI, QA, SCM, R&D, Warranty) targeting bearing failures across the G600/750 and G1150 gearbox families in commercial vehicle transmissions
- Achieved ~50% reduction in IPTV/EPV for bearing failures (24-MIS warranty window) through systematic root-cause analysis, supplier process audits, incoming inspection protocol upgrades, and assembly control improvements
- Realized annual warranty savings of ₹4.53 Cr for the LKP2518 family of heavy commercial tipper trucks
- Applied DoE, Gauge R&R, SPC (Cpk), ANOVA, and Ishikawa tools throughout the DMAIC cycle; authored full project documentation and presented to senior management

Operational Line Efficiency Improvement — G950 Gearbox Assembly Line

- Improved Overall Line Efficiency (OLE) of the G950 gearbox assembly line by ~12% through systematic takt-time re-balancing and elimination/merging of low-performing bottleneck stations
- Deployed PFMEA-driven quality control checkpoints at defect-generating stations, achieving a measurable reduction in in-process defects
- Conducted RULA (Rapid Upper Limb Assessment) ergonomic assessments at operator-intensive stations; redesigned tooling and fixture heights to reduce operator fatigue and associated defect rates
- Implemented a low-cost IoT-based ANDON monitoring system integrating with existing line PLCs to capture real-time assembly losses; deployed Ishikawa and 5-Why corrective action plans for identified loss causes

Value Engineering

- Case study on hydraulic tipping kit of heavy commercial tippers: alternate sourcing and design modifications for gear pumps; documented cost-saving roadmap
- Case study on G1150 (9-speed gearbox) synchronizing parts: application of new material/surface treatment to improve cone performance and durability; reduced warranty cost for synchronizing-part failures

Assistant Manager, New Product Introduction (NPI) — Transmissions
2016 – 2019

Tata Motors Ltd., CVBU, Jamshedpur, Jharkhand, India

TATA Kestrel 8×8— Auxiliary Gearbox Launch

- Led NPI for the auxiliary transmission (transfer case / auxiliary gearbox) for the TATA Kestrel 8×8, a 4th-generation amphibious infantry combat vehicle
- Managed full product lifecycle from design review through proto build, supplier development, DFA/DFM analysis, incoming quality inspection, assembly, integration testing, and standardization; achieved 100% delivery adherence at each proto stage
- Developed and maintained testing matrix for gearbox performance, NVH tests, and durability; coordinated with R&D, Supplier Quality, and Army acceptance teams for technical sign-off
- Integrated auxiliary gearbox into Product Lifecycle Management (PLM) system; managed engineering change management throughout development

New Gearbox Launches — Commercial & Defense

- Successfully launched 10 new gearbox variants (including 6x6 and 8x8 defense truck transmissions) from development stage into mass production
- Managed supplier development and qualification for new auxiliary gearbox components; conducted first-article inspections, dimensional audits, and process capability (Cpk) assessments
- Maintained 100% on-time production adherence throughout the NPI period despite parallel launches across multiple gearbox families

PUBLICATIONS

Journal Articles — First Author

[1] **Subudhi, S.K.**, Zhao, B., Wang, X., Ting, J., Takeuchi, I., Dasgupta, A., and Das, S. (2024). Flexible and twistable free-standing PDMS–magnetic-nanoparticle-based soft magnetic films with robust magnetic properties. *Flex. Print. Electron.*, 9(1), 015013. https://doi.org/10.1088/2058-8585/ad2480

[2] **Subudhi, S.K.**, Chandel, G.R., Sivasankar, V., and Das, S. (2024). Magnetic nanoparticle aggregation and complete de-encapsulation of such aggregates from a liquid drop interior. *ACS Appl. Mater. Interfaces*, 16(46), 64253–64263. https://doi.org/10.1021/acsami.4c14135

[3] **Subudhi, S.K.** and Das, S. (2023). Reliability of lab-on-a-chip technologies for wearable electronics: a perspective. Frontiers in Sensors, 4, 1283402. https://doi.org/10.3389/fsens.2023.1283402

[4] **Subudhi, S.K.**, Madhira, S.A., and Sas, S. (2026). Drop-to-Drop Transfer of Magnetic Activity: Imparting Significant Heterogeneity and Reactivity to a Drop via Functionality Transfer from Another Drop. *ACS Appl. Mater. Interfaces*. https://doi.org/10.1021/acsami.6c13139

Journal Articles — Contributing Author

[5] Zhao, B., Sivasankar, V., **Subudhi, S.K.**, Sinha, S., Nikfarjam, S., Woehl, T., and Das, S. (2023). Three-dimensional deposits from drying particle-laden drops. Langmuir, 39(28), 9773–9784. https://doi.org/10.1021/acs.langmuir.3c00768

[6] Zhao, B., Sivasankar, V.S., **Subudhi, S.K.**, Dasgupta, S., and Das, S. (2023). Printed carbon nanotube-based humidity sensors deployable on surfaces of widely varying curvatures. ACS Applied Nano Materials, 6(2), 1459–1474. https://doi.org/10.1021/acsanm.2c04873

[7] Zhao, B., Sivasankar, V.S., **Subudhi, S.K.**, Sinha, S., Dasgupta, A., and Das, S. (2022). Applications, fluid mechanics, and colloidal science of carbon-nanotube-based 3D-printable inks. Nanoscale, 14(40), 14858–14894. https://doi.org/10.1039/D2NR04151F

[8] Dhamsania, H.J., et al. [**incl. Subudhi, S.K.**] (2022). Physically soft magnetic films and devices: fabrication, properties, printability, and applications. Journal of Materials Chemistry C, 10(17), 6563–6589. https://doi.org/10.1039/D1TC05500G

[9] Zhao, B. et al. [**incl. Subudhi, S.K.**] (2026). Fabrication, Characterization, and Reliability Analysis of 3D Printed High Frequency, Wideband, and Flexible Carbon-Nanotube and Graphene-Oxide Based Rectangular Patch Antennas Deployable on Curvilinear and Deformable Surfaces. Accepted for publishing in *Frontiers in Antennas and Propagation.*

Conferences – Contributing Author

[10] Janamanchi, P., **Subudhi, S.**, Dasgupta, A., and Quinn, E. (2025). Thermal Cyclic Fatigue Reliability of Printed Hybrid Electronics (PHE). *2025 26th International Conference on Thermal, Mechanical and Multi-Physics Simulation and Experiments in Microelectronics and Microsystems (EuroSimE)*, Utrecht, Netherlands, 2025, pp. 1-4, doi: 10.1109/EuroSimE65125.2025.11006613.

[11] Zhao, B. *et al*., "3D Printed Wideband Flexible Carbon-Nanotube and Graphene-Oxide Based Rectangular Patch Antennas Deployable on Deformable Surfaces," *2026 IEEE International Conference on Flexible and Printable Sensors and Systems (FLEPS)*, Atlanta, GA, USA, 2026, pp. 1-4, doi: 10.1109/FLEPS69128.2026.11645063

TECHNICAL SKILLS

Additive Manufacturing & Fabrication

Direct-Write Printing: Extrusion printing (nScryptPro, Hyrel3D), Inks: Silver pastes, dielectric inks, CNT-GO ink formulation and characterization, Polymers (PDMS, PLA)

3D Printing: FDM (Ultimaker, Hyrel, CraftPro, Prusa & Formlabs platforms), SLA/DLP resin printing; print parameter optimization for dimensional accuracy

Subtractive & Traditional: CNC milling/turning, laser cutting/tooling, mold development, injection molding concepts

CAM Software: Autodesk Fusion 360, SolidWorks CAM

Computer-Aided Design & Simulation

CAD: SolidWorks (advanced — assemblies, drawings, simulation), Creo Parametric (Pro-E), AutoCAD

FEA / CAE: ANSYS Mechanical, ANSYS Workbench, ANSYS Icepak (thermal), Abaqus CAE (non-linear, visco-plasticity, creep)
CFD & Multiphysics: ANSYS Fluent, COMSOL Multiphysics (fluid-structure, electromagnetic, heat transfer), python-based Multiphysics Particle Simulator

Reliability Engineering & Testing

Accelerated Testing: Thermal cycling (JEDEC JESD22-A104), temperature-humidity bias (THB, JESD22-A101), vibration/shock testing (GEVS STD-7000B), deformational loading (most standards)
Failure Analysis: SEM/EDS fractography, optical cross-section microscopy, resistance monitoring, Physics of Failure (PoF) modelling, Weibull analysis, acceleration factor estimation
Quality & Process: Six Sigma (DMAIC), DoE (full factorial, RSM), PFMEA, DFA/DFM, ANOVA, SPC (Cpk), MSA / Gauge R&R, NPI process, PLM, BOM management

Materials Characterization

Thermal: TMA (CTE measurement), DSC (Tg, melting, crystallization), TGA (decomposition), DMA (storage modulus, loss factor, creep)
Microscopy & Surface: SEM (morphology, cross-section), EDS (elemental mapping), optical microscopy, laser scanning confocal microscopy, surface profilometry (stylus, white-light interferometry)
Rheology (inks & fluids): Viscosity (cone-plate, parallel-plate rheometry), surface tension (pendant drop), zeta-potential (DLS), particle size distribution (DLS, SEM-based)
Magnetic: VSM (magnetization curves, saturation magnetization, coercivity), hysteresis loop characterization
Dimensional: GD&T, CMM concepts, optical comparator, digital calipers/micrometers, MSA

Computation & Data Analysis

Scientific computing: MATLAB & Simulink (optimization toolbox, image processing toolbox, GUI development), Wolfram Mathematica, R / RStudio
Programming: Python (image processing, point tracking, data analysis), C, C++, Arduino IDE
Web & Scripting: HTML5, CSS3, JavaScript
Data & Reporting: JupyterLab, Microsoft Office (Excel, PowerPoint, Word), Power BI, Origin, Minitab

Instrumentation & Control

Control systems: Arduino (sensor integration, actuator control), MATLAB-based closed-loop control, stepper/servo motor control
Electronics: Circuit design (through-hole and SMD), PCB layout concepts, oscilloscope/multimeter/LCR meter, vector network analyser (VNA) operation
Sensors: Strain gauges, thermocouples, load cells, optical encoders, Hall-effect sensors, MEMS accelerometers

ACADEMIC PROJECTS

Synthesis & Applications of Reconfigurable Manipulator Assembly based on a 5-Bar Linkage

B.Tech. Thesis, 2016, NIT Rourkela
Advisor: Prof. J. Srinivas

- Synthesized and fabricated a 5-bar linkage reconfigurable manipulator with switchable end-effector trajectories
- Performed kinematic and dynamic analysis (forward/inverse kinematics, Jacobian, workspace mapping) in MATLAB
- Integrated closed-loop motor control via Arduino; developed a MATLAB GUI for user-friendly trajectory input, real-time visualization, and path optimization

Optimization of Taper Angle of Structures Machined by EDM Processes

Undergraduate Project, 2015–16, NIT Rourkela
Advisor: Prof. S. Gangopadhyay

- Applied Response Surface Methodology (Box-Behnken design) to optimize Material Removal Rate (MRR) and taper angle in reverse electro-discharge machining (R-EDM)
- Performed ANOVA to quantify the effects and interactions of current, pulse-on time, and electrode rotation speed; developed regression model with $R^2 > 0.95$

Combustion Modelling — Diesel Engine on Carbon Black Slurry

Undergraduate Project, 2015, NIT Rourkela
Advisor: Prof. S. Murugan

- Developed a two-zone thermodynamic combustion model for a DI diesel engine running on carbon black–water–diesel slurry fuel
- Predicted in-cylinder pressure, temperature, heat release rate, and emissions as functions of blend ratio; validated against experimental cylinder pressure traces

Electric Karting Vehicle — Chassis & Structural Design

Team Project, 2014–15

- Designed and fabricated chassis for a single-seat electric kart; performed FEA (Abaqus) for structural deformation and stress under design loads
- Served as mechanical designer for drive train integration; validated working model against design constraints

PRESENTATIONS & PROFESSIONAL SERVICE

Conference & Symposium Presentations

- Subudhi, S.K. Capillary Force Driven Extraction of MNPs in High-Viscous Mediums. Presented at GWU Symposium, Washington D.C. (2025)
- Subudhi, S.K. et al. Soft magnetic PDMS–nanoparticle films: fabrication and characterization. Presented at UMD GRAD, College Park, MD (2023)
- Cost-Optimization Case Study on a Hydraulic Tipping Kit. 35th INVEST Conference, Kolkata, IN, Nov 2019.

Professional & Academic Leadership Roles

- ENME Representative, Graduate Student Government (GSG), University of Maryland — Jul 2022 – May 2025
  - Elected departmental representative; served as Chair of Academic Affairs Committee, advocating for graduate student academic policy at the university level
- Executive Body Member, Mechanical & Reliability Engineering Graduate Association (MEGA), University of Maryland — Jul 2024 – Present
- Cross-functional team leader (3–5 engineers), Tata Motors Ltd. — 2018–2020
- Assembly Team Leader (2-3 floor managers, 30-40 operators), Tata Motors ltd. — 2019–2020

WORK AUTHORIZATION

F-1 student visa holder (India). Eligible for 12-month OPT + 24-month STEM OPT extension upon graduation, providing up to 36 months of US work authorization. Employer requires E-Verify enrollment only — no H-1B petition, lottery, or attorney fees required for the initial employment period. Available to start mid-2026.

REFERENCES

Available upon request. Primary academic reference: Prof. Siddhartha Das, Department of Mechanical Engineering, University of Maryland, College Park (sidd@umd.edu).